

Individual Assessment Coversheet

To be attached to the front of the assessment.

Campus: Tyger Valley

Faculty: Information Technology

Module Code: ITMBA2-33

Group: 3

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Indicate	Yes	No
Plagiarism report attached		

Declaration:

I declare that this assessment is my own original work except for source material explicitly acknowledged. I also declare that this assessment or any other of my original work related to it has not been previously, or is not being simultaneously, submitted for this or any other course. I am aware of the AI policy and acknowledge that I have not used any AI technology to generate or manipulate data, other than as permitted by the assessment instructions. I also declare that I am aware of the Institution's policy and regulations on honesty in academic work as set out in the Conditions of Enrolment, and of the disciplinary guidelines applicable to breaches of such policy and regulations.

Signature 	Date 29/08/2025
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Lecturer's Comments:

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Signature	Date
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Section A

Deliverable 1

1.1) The Software Development Life Cycle (SDLC) that can also be known as Program Development Life Cycle (PDLC) is a process, made up of individual phases, that an organisation can follow when developing programs. These phases include requirements gathering, planning and designing the program as well as building, testing and fully implementing the program (KIELY, Michael.T., 2015). Each of these identified phases can assist in the creation of the TripBuddy application.

The requirement gathering and analysis phase requires developers to identify and define what the user desires from the TripBuddy application to improve customer satisfaction and engagement. The quality of the TripBuddy application is determined by how engaged the user is to the personalisation aspects provided by the program, therefore the developers must ensure that they fully understand how users would like to customise their trip memories before they start developing the application. The user's requirements could be obtained through communication methods such as questionnaires and surveys that the development team could analyse (Sharma, 2017). A good understanding of the user's requirements can lead to a high-quality application that retains the user's interest in TripBuddy and promotes brand loyalty. However, failing to meet these requirements can lead to a decrease in user engagement that can lead to the application failing.

Furthermore, the planning and designing phase will make use of the TripBuddy functionality outlined in beginning as well as the requirements that were received by the users to plan how the low-level component and algorithm issues should be addressed and determine how the architectural view of the TripBuddy application should be designed (Jain, 2011). Ensuring that all aspects that should be implemented into the TripBuddy application are planned for in advance enables the development team to identify potential issues and conflicts between the functionality and user requirements. Identifying these issues in the early stages of the PDLC allows the developers to design a solution to neutralise these issues, enabling the development team to spend time improving TripBuddy's user experience instead of solving issues that were left unchecked. In addition, designing an appropriate architectural view for TripBuddy in a manner where the required functionality can be interacted with easily and aesthetics is a focus point, can greatly improve the user experience. The user experience is improved due to the ease of use and style of the application that makes it more memorable for user, which can improve brand loyalty leading to a successful application.

Moreover, the TripBuddy application should then be developed by following the guidelines set during the planning and designing phase and tested to ensure that there are no issues that went undiscovered during the planning and designing phase or introduced by mistakes made when the application was developed. Ensuring that any issues are resolved can greatly improve the user experience, because application issues could decrease the overall enjoyment of the user when they use TripBuddy. It can also result in a decrease in brand loyalty as users could consider the application to be low quality depending on the severity of issues that user encounter such as an issue that makes the whole application slower or an issue that results in the user's trip memories to be deleted or corrupted. After the TripBuddy application is fully developed and tested, it can then be fully implemented in a manner where users are able to provide feedback and report issues that they have discovered. This provides the development team the opportunity to plan for further

improvements to the TripBuddy application based on the feedback provided by users. This can improve the user experience for new users who recently started using the application as well as improves brand loyalty as the application continues to improve over a long period of time.

Deliverable 2

AndroidManifest.xml file:

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/trip_buddy_icon"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/trip_buddy_icon_round"
        android:supportRtl="true"
        android:theme="@style/Theme.TripBuddy">
        <activity
            android:name=".CreateMemoryScreen"
            android:exported="false" />
        <activity
            android:name=".SettingsScreen"
            android:exported="false" />
        <activity
            android:name=".UserLoginScreen"
            android:exported="false" />
        <activity
            android:name=".Gallery"
            android:exported="false" />
        <activity
            android:name=".PlanTripScreen"
            android:exported="false" />
        <activity
            android:name=".HomeDashboard"
            android:exported="false" />
        <activity
            android:name=".SplashScreen"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

trip_buddy_icon_round.xml file:

```
<?xml version="1.0" encoding="utf-8"?>
<adaptive-icon xmlns:android="http://schemas.android.com/apk/res/android">
    <background android:drawable="@color/trip_buddy_icon_background"/>
```

```
<foreground android:drawable="@drawable/trip_buddy_icon_foreground"/>
</adaptive-icon>
```

trip_buddy_icon.xml file:

```
<?xml version="1.0" encoding="utf-8"?>
<adaptive-icon xmlns:android="http://schemas.android.com/apk/res/android">
    <background android:drawable="@color/trip_buddy_icon_background"/>
    <foreground android:drawable="@drawable/trip_buddy_icon_foreground"/>
</adaptive-icon>
```

Trip_buddy_icon_foreground.xml file:

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="108dp"
    android:height="108dp"
    android:viewportWidth="960"
    android:viewportHeight="960"
    android:tint="#008254">
    <group android:scaleX="0.58"
        android:scaleY="0.58"
        android:translateX="201.6"
        android:translateY="201.6">
        <path
            android:fillColor="@android:color/white"
            android:pathData="M240,880Q207,880 183.5,856.5Q160,833 160,800L160,320Q160,264
194,222Q228,180 280,166L280,80L400,80L400,160L560,160L560,80L680,80L680,166Q732,180
766,222Q800,264 800,320L800,800Q800,833 776.5,856.5Q753,880
720,880L240,880ZM240,800L720,800Q720,800 720,800Q720,800 720,800L720,320Q720,287
696.5,263.5Q673,240 640,240L320,240Q287,240 263.5,263.5Q240,287 240,320L240,800Q240,800
240,800ZM580,640L660,640L660,480L300,480L300,560L580,560L580,640ZM480,520Q480,520
480,520L480,520Q480,520 480,520Q480,520 480,520L480,520Q480,520 480,520Z"/>
        </group>
    </vector>
```

trip_buddy_icon_background.xml file:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="trip_buddy_icon_background">#8EE3AB</color>
</resources>
```

activity_splash_screen.xml file:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">
```

```

    android:background="@color/black"
    tools:context=".SplashScreen">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center"
        android:textSize="70dp"
        android:layout_marginTop="40dp"
        android:textStyle="italic"
        android:textColor="@color/white"
        android:text="Trip Buddy"
    />

    <ImageView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:src="@drawable/trip_buddy_logo"
        android:layout_marginTop="5dp"
        android:id="@+id/ivLogo"
        android:layout_gravity="center"
    />
</LinearLayout>

```

SplashScreen.java file:

```

package com.example.tripbuddy;

import android.content.Intent;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.os.Handler;
import android.os.Looper;
import android.view.animation.Animation;
import android.view.animation.AnimationUtils;
import android.widget.ImageView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class SplashScreen extends AppCompatActivity {
    TripDetails_SQLite dbTrip;
    MemoryDetails_SQLite dbMemory;

    private ImageView ivLogo;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        dbTrip = new TripDetails_SQLite(this);
        dbMemory = new MemoryDetails_SQLite(this);
        dbTrip.deleteAllTripEntries();
        dbMemory.deleteAllMemoryEntries();
    }
}

```

```

        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_splash_screen);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets)
-> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });
        ivLogo = findViewById(R.id.ivLogo);
        Animation anim = AnimationUtils.loadAnimation(getApplicationContext(),
R.anim.fade_in_anim);
        ivLogo.setAnimation(anim);

        SharedPreferences prefs = getSharedPreferences(CurrentUserDetails.PREFS_NAME,
MODE_PRIVATE);
        boolean loggedIn = prefs.getBoolean(CurrentUserDetails.KEY_LOGGEDIN, false);

        if (loggedIn == true) {
CurrentUserDetails.populateUserDetails(prefs.getString(CurrentUserDetails.KEY_USERNAME,
""), prefs.getInt(CurrentUserDetails.KEY_TRIPAMOUNT, 0),
prefs.getBoolean(CurrentUserDetails.KEY_DARKMODE, false));
            new Handler().postDelayed(() -> {
                if (CurrentUserDetails.darkMode == true){
AppCompatActivity.setDefaultNightMode(AppCompatActivity.MODE_NIGHT_YES);
                }
                Intent intent = new Intent(SplashScreen.this, HomeDashboard.class);
                startActivity(intent);
            }, 5000);
        }
        else{
            new Handler().postDelayed(() -> {
                Intent intent = new Intent(SplashScreen.this, UserLoginScreen.class);
                startActivity(intent);
            }, 5000);
        }
    }
}

```

colors.xml file:

```

<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFF</color>
</resources>

```

dashboard_home.xml file:

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"

```



```

xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:padding="16dp"
tools:context=".HomeDashboard">

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:padding="10dp"
    android:text="TripBuddy Home Dashboard"
    android:textColor="@color/white"
    android:textAlignment="center"
    android:textSize="20dp"
    android:textStyle="bold"
    android:background="@color/cardview_dark_background"
/>

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/txtUsername"
    android:padding="10dp"
    android:text="Username: keenan@gmail.com"
    android:layout_marginTop="60dp"
    android:textSize="12dp"
    android:textStyle="bold"
/>

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/txtWelcomeMessage"
    android:padding="10dp"
    android:text="Welcome to the TripBuddy Home Dashboard"
    android:textAlignment="center"
    android:layout_marginTop="120dp"
    android:textSize="15dp"
    android:textStyle="bold"
/>

<RelativeLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_centerInParent="true"
    >
    <Button
        android:layout_width="250dp"
        android:layout_height="wrap_content"
        android:layout_centerHorizontal="true"
        android:text="View Gallery"
        android:id="@+id/btnViewGallery"
        />

```

```

        <Button
            android:layout_width="250dp"
            android:layout_height="wrap_content"
            android:layout_centerHorizontal="true"
            android:text="Plan Trip"
            android:id="@+id/btnPlanTrip"
            android:layout_marginTop="50dp"
        />

        <Button
            android:layout_width="250dp"
            android:layout_height="wrap_content"
            android:layout_centerHorizontal="true"
            android:text="Create Memory"
            android:id="@+id/btnCreateMemory"
            android:layout_marginTop="100dp"
        />

        <Button
            android:layout_width="250dp"
            android:layout_height="wrap_content"
            android:layout_centerHorizontal="true"
            android:text="Settings"
            android:layout_marginTop="150dp"
            android:id="@+id/btnSettings"
        />
    </RelativeLayout>

    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Log off"
        android:layout_centerHorizontal="true"
        android:layout_alignParentBottom="true"
        android:id="@+id/btnLogOff"
    />
</RelativeLayout>

```

HomeDashboard.java file:

```

package com.example.tripbuddy;

import android.content.Intent;
import android.content.SharedPreferences;
import android.database.Cursor;
import android.os.Bundle;
import android.os.Handler;
import android.transition.TransitionManager;
import android.view.MotionEvent;
import android.view.View;
import android.view.animation.Animation;
import android.view.animation.AnimationUtils;
import android.widget.Button;
import android.widget.TextView;

```

```

import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class HomeDashboard extends AppCompatActivity {
    private Button btnViewGallery, btnPlanTrip, btnCreateMemory, btnSettings, btnLogOff;
    private TextView txtWelcomeMessage, txtUsername;
    Animation scaleUpAnim, fadeOut;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.dashboard_home);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets)
-> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });
        txtWelcomeMessage = findViewById(R.id.txtWelcomeMessage);
        txtUsername = findViewById(R.id.txtUsername);

        btnViewGallery = findViewById(R.id.btnViewGallery);
        btnPlanTrip = findViewById(R.id.btnPlanTrip);
        btnCreateMemory = findViewById(R.id.btnCreateMemory);
        btnSettings = findViewById(R.id.btnSettings);
        btnLogOff = findViewById(R.id.btnLogOff);

        scaleUpAnim = AnimationUtils.loadAnimation(getApplicationContext(),
R.anim.size_up_anim);
        fadeOut = AnimationUtils.loadAnimation(getApplicationContext(),
R.anim.fade_out_anim);
        txtUsername.setText("Username: " + CurrentUserDetails.username);
        txtWelcomeMessage.setAnimation(fadeOut);
        new Handler().postDelayed(() -> {
            txtWelcomeMessage.setVisibility(View.INVISIBLE);
        }, 4500);

        btnViewGallery.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                viewMemoryGallery();
            }
        });

        btnCreateMemory.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                createNewMemory();
            }
        });
    }
}

```

```

});

btnSettings.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        changeSettings();
    }
});

btnLogOff.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        userLogOff();
    }
});

btnPlanTrip.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        planTripScreen();
    }
});
}

private void viewMemoryGallery() {
    btnViewGallery.startAnimation(scaleUpAnim);
    Intent intent = new Intent(HomeDashboard.this, Gallery.class);
    startActivity(intent);
}

private void createNewMemory() {
    btnCreateMemory.startAnimation(scaleUpAnim);
    Intent intent = new Intent(HomeDashboard.this, CreateMemoryScreen.class);
    startActivity(intent);
}

private void changeSettings() {
    btnSettings.startAnimation(scaleUpAnim);
    Intent intent = new Intent(HomeDashboard.this, SettingsScreen.class);
    startActivity(intent);
}

private void userLogOff() {
    btnLogOff.startAnimation(scaleUpAnim);
    SharedPreferences prefs = getSharedPreferences(CurrentUserDetails.PREFS_NAME,
MODE_PRIVATE);
    SharedPreferences.Editor editor = prefs.edit();
    editor.putBoolean(CurrentUserDetails.KEY_LOGGEDIN, true);
    editor.apply();

    CurrentUserDetails.userLogOff();
    Intent intent = new Intent(HomeDashboard.this, UserLoginScreen.class);
    startActivity(intent);
}

private void planTripScreen() {

```

```
        btnPlanTrip.startAnimation(scaleUpAnim);  
        Intent intent = new Intent(HomeDashboard.this, PlanTripScreen.class);  
        startActivity(intent);  
    }  
}
```

Deliverable 3

plan_trip_screen.xml file:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".PlanTripScreen">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:padding="10dp"
        android:text="TripBuddy Trip Planner"
        android:textColor="@color/white"
        android:textAlignment="center"
        android:textSize="20dp"
        android:textStyle="bold"
        android:background="@color/cardview_dark_background"
    />

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="50dp"
        android:text="Total trips planned: "
        android:id="@+id/txtTripsPlanned"
    />

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="75dp"
        android:text="Previous trip activities: "
        android:id="@+id/txtPrevTripActivities"
    />

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="100dp"
        android:text="Previous custom expenses: "
        android:id="@+id/txtPrevCustomExpenses"
    />

    <RelativeLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_centerInParent="true"
```

```

>
<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Destination"
    android:layout_marginTop="50dp"
    android:id="@+id/edtDestination"
/>
<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Notes"
    android:layout_marginTop="100dp"
    android:id="@+id/edtNote"
/>
<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Custom Expenses"
    android:layout_marginTop="150dp"
    android:id="@+id/edtCustomExpenses"
/>

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="215dp"
    android:text="Select one or more trip activities"
    android:id="@+id/txtTripActivities"
    android:drawableEnd="@android:drawable/arrow_down_float"
    android:background="@android:drawable/editbox_background"
/>
<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Meal Expenses"
    android:layout_marginTop="250dp"
    android:id="@+id/edtMealExpenses"
/>

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Subtotal:"
    android:layout_marginTop="325dp"
    android:id="@+id/txtSubtotal"
/>
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Discount:"
    android:layout_marginTop="350dp"
    android:id="@+id/txtDiscount"
/>
<TextView
    android:layout_width="match_parent"

```

```

        android:layout_height="wrap_content"
        android:text="Final Budget:"
        android:layout_marginTop="375dp"
        android:id="@+id/txtFinalBudget"
    />
    <Button
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Confirm Trip"
        android:layout_marginTop="420dp"
        android:id="@+id/btnConfirmTrip"
    />
</RelativeLayout>

<Button
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Return to Home Dashboard"
    android:layout_alignParentBottom="true"
    android:id="@+id/btnReturnHome"
/>

</RelativeLayout>

```

PlanTripScreen.java file:

```

package com.example.tripbuddy;

import android.content.DialogInterface;
import android.content.Intent;
import android.content.SharedPreferences;
import android.database.Cursor;
import android.os.Bundle;
import android.text.Editable;
import android.text.TextWatcher;
import android.view.View;
import android.view.animation.Animation;
import android.view.animation.AnimationUtils;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import java.util.ArrayList;

public class PlanTripScreen extends AppCompatActivity {
    TripDetails_SQLite dbTripDetails;

```



```

        private TextView txtFinalBudget, txtSubtotal, txtDiscount, txtTripActivities,
txtTripsPlanned, txtPrevTripActivities, txtPrevCustomExpenses;
        private EditText edtCustomExpenses, edtMealExpenses, edtDestination, edtNote;
        private Button btnConfirmTrip, btnReturnHome;

        boolean[] selectedActivities;
        ArrayList<Integer> activityList = new ArrayList<>();
        String[] tripActivities = {"Sightseeing(R1050.00)", "Hiking(R650.00)",
"Dining(R600.00)", "Museum tours(R800.00)"};
        Double[] tripActivityPrice = {1050.00, 650.00, 600.00, 800.00};
        Double subtotal = 0.00, discount = 0.00, finalBudget = 0.00, prevCustomExpenses =
0.00, prevMealExpenses = 0.00;
        Animation scaleUpAnim;

        @Override
        protected void onCreate(Bundle savedInstanceState) {
            dbTripDetails = new TripDetails_SQLite(this);
            super.onCreate(savedInstanceState);
            EdgeToEdge.enable(this);
            setContentView(R.layout.plan_trip_screen);
            ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets)
-> {
                Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
                v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
                return insets;
            });
            scaleUpAnim = AnimationUtils.loadAnimation(getApplicationContext(),
R.anim.size_up_anim);

            txtFinalBudget = findViewById(R.id.txtFinalBudget);
            txtSubtotal = findViewById(R.id.txtSubtotal);
            txtDiscount = findViewById(R.id.txtDiscount);
            txtTripActivities = findViewById(R.id.txtTripActivities);
            txtTripsPlanned = findViewById(R.id.txtTripsPlanned);
            txtPrevCustomExpenses = findViewById(R.id.txtPrevCustomExpenses);
            txtPrevTripActivities = findViewById(R.id.txtPrevTripActivities);
            edtCustomExpenses = findViewById(R.id.edtCustomExpenses);
            edtMealExpenses = findViewById(R.id.edtMealExpenses);
            edtDestination = findViewById(R.id.edtDestination);
            edtNote = findViewById(R.id.edtNote);
            btnConfirmTrip = findViewById(R.id.btnConfirmTrip);
            btnReturnHome = findViewById(R.id.btnReturnHome);

            selectedActivities = new boolean[tripActivities.length];
            txtTripsPlanned.setText("Total trips planned: " +
Integer.toString(CurrentUserDetails.totalTripsPlanned) + " trip(s)");

            txtTripActivities.setOnClickListener(new View.OnClickListener() {
                @Override
                public void onClick(View v) {
                    selectATripActivity(); (jangirkaran17, 2025)
                }
            });

            btnConfirmTrip.setOnClickListener(new View.OnClickListener() {

```

```

        @Override
        public void onClick(View v) {
            saveNewTrip();
        }
    });

    edtDestination.addTextChangedListener(new TextWatcher() {
        @Override
        public void afterTextChanged(Editable s) {

        }

        @Override
        public void beforeTextChanged(CharSequence s, int start, int count, int
after) {

        }

        @Override
        public void onTextChanged(CharSequence s, int start, int before, int count)
{
            Cursor cursor = dbTripDetails.getAllTripEntries();
            String destination = edtDestination.getText().toString().toLowerCase();
            String strActivities;
            String strExpenses;

            txtPrevTripActivities.setText("Previous trip activities:");
            txtPrevCustomExpenses.setText("Previous custom expenses:");
            if (cursor.moveToFirst()) {
                do {
                    if (cursor.getString(2).toLowerCase().equals(destination) &&
cursor.getString(1).equals(CurrentUserDetails.username)){
                        strActivities = ("Previous trip activities: " +
cursor.getString(3));
                        strExpenses = ("Previous custom expenses: R" +
String.format("%.2f", cursor.getDouble(4)));
                        txtPrevTripActivities.setText(strActivities);
                        txtPrevCustomExpenses.setText(strExpenses);
                        return;
                    }
                } while (cursor.moveToNext());
            }
            cursor.close();
        }
    });

    edtCustomExpenses.addTextChangedListener(new TextWatcher() { (Pirple, 2019)
        @Override
        public void afterTextChanged(Editable s) {

        }

        @Override
        public void beforeTextChanged(CharSequence s, int start, int count, int
after) {

```

```

    }

    @Override
    public void onTextChanged(CharSequence s, int start, int before, int count)
{
        String expense = s.toString().trim();

        if (expense.isEmpty()) {
            calculateBudget("0.00", "0.00", true);
        }
        else{
            calculateBudget(expense, "0.00", true);
        }
    }
});

edtMealExpenses.addTextChangedListener(new TextWatcher() {
    @Override
    public void afterTextChanged(Editable s) {

    }

    @Override
    public void beforeTextChanged(CharSequence s, int start, int count, int
after) {

    }

    @Override
    public void onTextChanged(CharSequence s, int start, int before, int count)
{
        String expense = s.toString().trim();

        if (expense.isEmpty()) {
            calculateBudget("0.00", "0.00", false);
        }
        else{
            calculateBudget("0.00", expense, false);
        }
    }
});

btnReturnHome.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        returnHome();
    }
});

}

private void saveNewTrip() {
    btnConfirmTrip.startAnimation(scaleUpAnim);

    Double customEx = 0.00;
    Double mealEx = 0.00;
    String destination = edtDestination.getText().toString();

```

```

String note = edtNote.getText().toString();
String activities = txtTripActivities.getText().toString();

try {
    customEx = Double.parseDouble(edtCustomExpenses.getText().toString());
    mealEx = Double.parseDouble(edtMealExpenses.getText().toString());

    if (!destination.isBlank() && !destination.isEmpty() && !note.isBlank()
&& !note.isEmpty() && !activityList.isEmpty()) {
        SharedPreferences prefs =
getSharedPreferences(CurrentUserDetails.PREFS_NAME, MODE_PRIVATE);
        SharedPreferences.Editor editor = prefs.edit();
        editor.putInt(CurrentUserDetails.KEY_TRIPAMOUNT,
CurrentUserDetails.totalTripsPlanned++);
        editor.apply();
        txtTripsPlanned.setText("Total trips planned: " +
Integer.toString(CurrentUserDetails.totalTripsPlanned) + " trip(s)");
        boolean trip = dbTripDetails.addNewTrip(destination, activities,
customEx);
        if (trip == true) {
            Toast.makeText(this, "You have successfully created a new trip!",
Toast.LENGTH_SHORT).show();
        }
        else{
            Toast.makeText(this, "Error! You must enter information in the
destination, note and activity fields.", Toast.LENGTH_SHORT).show();
        }
    }
    catch(Exception e){
        Toast.makeText(this, "Error! Your expenses MUST NOT contain any letters.",
Toast.LENGTH_SHORT).show();
    }
}

private void selectATripActivity() {
    AlertDialog.Builder builder = new AlertDialog.Builder(PlanTripScreen.this);
    builder.setTitle("Select one or more trip activities");
    builder.setCancelable(false);

    builder.setMultiChoiceItems(tripActivities, selectedActivities, new
DialogInterface.OnMultiChoiceClickListener() {
        @Override
        public void onClick(DialogInterface dialog, int activity, boolean isChecked)
{
            if (isChecked){
                activityList.add(activity);
                subtotal += tripActivityPrice[activity];
            }
            else{
                activityList.remove(Integer.valueOf(activity));
                subtotal -= tripActivityPrice[activity];
            }
        }
    });
}

```

```

        builder.setPositiveButton("Confirm", new DialogInterface.OnClickListener() {
            @Override
            public void onClick(DialogInterface dialog, int which) {
                StringBuilder stringBuilder = new StringBuilder();
                for (int i = 0; i < activityList.size(); i++){
                    stringBuilder.append(tripActivities[activityList.get(i)]);
                    if (i != (activityList.size()-1)){
                        stringBuilder.append(", ");
                    }
                }
                txtTripActivities.setText(stringBuilder.toString());
                printSubtotal();
            }
        });
        builder.show();
    }

    private void calculateBudget(String CustomExpense, String MealExpense, boolean
cusExpense) {
        Double customExpense = 0.00;
        Double mealExpense = 0.00;

        try{
            customExpense = Double.parseDouble(CustomExpense);
            mealExpense = Double.parseDouble(MealExpense);

            if (customExpense >= 0 && mealExpense >= 0) {
                if (cusExpense == true) {
                    subtotal -= prevCustomExpenses;
                }
                else {
                    subtotal -= prevMealExpenses;
                }
                subtotal += (customExpense + mealExpense);
                if (cusExpense == true) {
                    prevCustomExpenses = customExpense;
                }
                else {
                    prevMealExpenses = mealExpense;
                }
                printSubtotal();
            }
            else{
                Toast.makeText(this, "Error! Your values MUST NOT be negative.",
Toast.LENGTH_SHORT).show();
            }
        }
        catch(Exception e){
            Toast.makeText(this, "Error! Your expenses MUST NOT contain any letters.",
Toast.LENGTH_SHORT).show();
        }
    }

    private void printSubtotal(){
        if (CurrentUserDetails.totalTripsPlanned >= 3){
            discount = (subtotal * 0.1);
        }
    }

```

```

    }
    else{
        discount = 0.00;
    }
    finalBudget = (subtotal - discount);

    String strSubTotal = String.format("%.2f", subtotal);
    String strDiscount = String.format("%.2f", discount);
    String strBudget = String.format("%.2f", finalBudget);
    txtSubtotal.setText("Subtotal: R" + strSubTotal);
    txtDiscount.setText("Discount: R" + strDiscount);
    txtFinalBudget.setText("Final Budget: R" + strBudget);
}

private void returnHome() {
    btnReturnHome.startAnimation(scaleUpAnim);
    Intent intent = new Intent(PlanTripScreen.this, HomeDashboard.class);
    startActivity(intent);
}
}

```

Deliverable 4

fade_in_anim.xml file:

```
<?xml version="1.0" encoding="utf-8"?>
<set xmlns:android="http://schemas.android.com/apk/res/android"
    android:interpolator="@android:anim/accelerate_interpolator"
    android:duration="4500">

    <alpha
        android:fromAlpha="0.0"
        android:toAlpha="1.0"
        /> (Coding, 2020)

</set>
```

fade_out_anim.xml file:

```
<?xml version="1.0" encoding="utf-8"?>
<set xmlns:android="http://schemas.android.com/apk/res/android"
    android:interpolator="@android:anim/accelerate_interpolator"
    android:duration="4500">

    <alpha
        android:fromAlpha="1.0"
        android:toAlpha="0.0"
        />

</set>
```

size_up_anim.xml file:

```
<?xml version="1.0" encoding="utf-8"?>
<scale xmlns:android="http://schemas.android.com/apk/res/android"
    android:fromXScale="1.2"
    android:fromYScale="1.2"
    android:toXScale="1.0"
    android:toYScale="1.0"
    android:pivotX="50%"
    android:pivotY="50%"
    android:duration="100"
    > (Guru, 2020)

</scale>
```

MemoryDetails_SQLite.java class file:

```
package com.example.tripbuddy;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
```

```

import android.database.sqlite.SQLiteOpenHelper;

public class MemoryDetails_SQLite extends SQLiteOpenHelper {
    private static final String DATABASE_NAME = "memory.db";
    private static final int DATABASE_VERSION = 1;
    public static final String TABLE_DETAILS = "details";
    public static final String KEY_ID = "memory_id";
    public static final String KEY_USERNAME = "username";
    public static final String KEY_DESCRIPTION = "description";
    public static final String KEY_IMAGE = "image";
    public static final String KEY_MUSIC = "music_track";

    public MemoryDetails_SQLite(Context context) {
        super(context, DATABASE_NAME, null, DATABASE_VERSION);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        String CREATE_TABLE = "CREATE TABLE " + TABLE_DETAILS + "("
            + KEY_ID + " INTEGER PRIMARY KEY AUTOINCREMENT,"
            + KEY_USERNAME + " TEXT,"
            + KEY_DESCRIPTION + " TEXT,"
            + KEY_IMAGE + " TEXT,"
            + KEY_MUSIC + " INTEGER " + ")";
        db.execSQL(CREATE_TABLE);
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_DETAILS);
        onCreate(db);
    }

    public boolean createNewMemory(String description, String image, int music) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues values = new ContentValues();
        values.put(KEY_USERNAME, CurrentUserDetails.username);
        values.put(KEY_DESCRIPTION, description);
        values.put(KEY_IMAGE, image);
        values.put(KEY_MUSIC, music);
        long num = db.insert(TABLE_DETAILS, null, values); (CodingWithMitch, 2017)
        db.close();
        if (num == -1){
            return false;
        }
        else{
            return true;
        }
    }

    public Cursor getAllMemoryEntries() {
        SQLiteDatabase db = this.getReadableDatabase();
        return db.rawQuery("SELECT * FROM " + TABLE_DETAILS, null);
    }

    public void deleteAllMemoryEntries() {

```



```

        SQLiteDatabase db = this.getWritableDatabase();
        db.delete(TABLE_DETAILS, null, null);
        db.close();
    }
}

```

activity_create_memory_screen.xml file:

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".CreateMemoryScreen">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:padding="10dp"
        android:text="TripBuddy Memory Creation"
        android:textColor="@color/white"
        android:textAlignment="center"
        android:textSize="20dp"
        android:textStyle="bold"
        android:background="@color/cardview_dark_background"
    />

    <EditText
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter memory description"
        android:layout_marginTop="60dp"
        android:id="@+id/edtMemoryDescription"
    />

    <ImageView
        android:layout_width="200dp"
        android:layout_height="200dp"
        android:layout_marginTop="110dp"
        android:id="@+id/ivPicture"
        android:layout_centerHorizontal="true"
    />

    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="320dp"
        android:layout_centerHorizontal="true"
        android:textSize="18dp"
        android:text="Insert image"
        android:id="@+id/btnInsertImage"
    />

```

```

/>

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="400dp"
    android:textSize="18dp"
    android:text="Select your music track:"
/>

<RadioGroup
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="420dp"
    android:id="@+id/rgSongTrack"
>
    <RadioButton
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Song 1"
        android:id="@+id/rbSong1"
    />
    <RadioButton
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Song 2"
        android:id="@+id/rbSong2"
    />
    <RadioButton
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Song 3"
        android:id="@+id/rbSong3"
    />
    <RadioButton
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Song 4"
        android:id="@+id/rbSong4"
    />
</RadioGroup>

<Button
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_centerHorizontal="true"
    android:layout_alignParentBottom="true"
    android:layout_marginBottom="5dp"
    android:layout_alignParentLeft="true"
    android:text="Save memory"
    android:id="@+id/btnSaveMemory"
/>

<Button
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"

```

```

        android:layout_centerHorizontal="true"
        android:layout_alignParentBottom="true"
        android:layout_alignParentRight="true"
        android:layout_marginBottom="5dp"
        android:text="Return to Home Dashboard"
        android:id="@+id/btnReturnHome"
    />

```

</RelativeLayout>

CreateMemoryScreen.java file:

```

package com.example.tripbuddy;

import android.content.Context;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.view.animation.Animation;
import android.view.animation.AnimationUtils;
import android.widget.Button;
import android.widget.EditText;
import android.widget.ExpandableListView;
import android.widget.ImageView;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import java.io.File;
import java.io.FileNotFoundException;
import java.io.FileOutputStream;
import java.io.InputStream;

public class CreateMemoryScreen extends AppCompatActivity {
    MemoryDetails_SQLite dbMemoryDetails;
    private Button btnSaveMemory, btnInsertImage, btnReturnHome;
    private EditText edtMemoryDescription;
    private ImageView ivPicture;
    private RadioGroup rgSongTrack;
    private RadioButton rbSong1;
    private String imagePath;
    private boolean imageSelected = false, songSelected = false;
    private Integer songPath;
    private Uri uriImage;
    Animation scaleUpAnim;

```

```

@Override
protected void onCreate(Bundle savedInstanceState) {
    dbMemoryDetails = new MemoryDetails_SQLite(this);
    songPath = R.raw.song1;
    imageSelected = false;
    songSelected = false;
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_create_memory_screen);
    ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets)
-> {
        Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
        v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
        return insets;
    });
    scaleUpAnim = AnimationUtils.loadAnimation(getApplicationContext(),
R.anim.size_up_anim);
    rbSong1 = findViewById(R.id.rbSong1);
    rgSongTrack = findViewById(R.id.rgSongTrack);
    ivPicture = findViewById(R.id.ivPicture);
    btnSaveMemory = findViewById(R.id.btnSaveMemory);
    btnReturnHome = findViewById(R.id.btnReturnHome);
    btnInsertImage = findViewById(R.id.btnInsertImage);
    edtMemoryDescription = findViewById(R.id.edtMemoryDescription);

    btnReturnHome.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            btnReturnHome.startAnimation(scaleUpAnim);
            returnToHomeDashboard();
        }
    });

    rgSongTrack.setOnCheckedChangeListener(new RadioGroup.OnCheckedChangeListener()
{
    @Override
    public void onCheckedChanged(@NonNull RadioGroup group, int checkedId) {
        if (checkedId == R.id.rbSong1){
            songPath = R.raw.song1;
            songSelected = true;
        }
        if (checkedId == R.id.rbSong2){
            songPath = R.raw.song2;
            songSelected = true;
        }
        if (checkedId == R.id.rbSong3){
            songPath = R.raw.song3;
            songSelected = true;
        }
        if (checkedId == R.id.rbSong4){
            songPath = R.raw.song4;
            songSelected = true;
        }
    }
});
}

```

```

        btnInsertImage.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                insertUserImage(); (Haroon, 2018)
            }
        });

        btnSaveMemory.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                saveMemory();
            }
        });
    }

    private void returnToHomeDashboard() {
        Intent intent = new Intent(CreateMemoryScreen.this, HomeDashboard.class);
        startActivity(intent);
    }

    private void saveMemory() {
        btnSaveMemory.startAnimation(scaleUpAnim);
        String description = edtMemoryDescription.getText().toString();

        if (imageSelected == true) {
            if (description.isBlank() || description.isEmpty()) {
                Toast.makeText(this, "Error! Please enter the description of your
memory.", Toast.LENGTH_SHORT).show();
            }
            else {
                if (songSelected == true) {
                    try {
                        boolean memorySaved =
dbMemoryDetails.createNewMemory(description, uriImage.toString(), songPath);

                        if (memorySaved == true) {
                            Toast.makeText(this, "Your memory has been created and
saved successfully.", Toast.LENGTH_SHORT).show();
                            returnToHomeDashboard();
                        }
                        else{
                            Toast.makeText(this, "Error! Your memory could not be
saved.", Toast.LENGTH_SHORT).show();
                        }
                    } catch (Exception e) {
                        Toast.makeText(this, "Error! File not found.",
Toast.LENGTH_SHORT).show();
                    }
                }
                else {
                    Toast.makeText(this, "Error! Please select a song for your memory.",
Toast.LENGTH_SHORT).show();
                }
            }
        }
    }
}

```

```

        else{
            Toast.makeText(this, "Error! Please insert an image for you memory.",
Toast.LENGTH_SHORT).show();
        }
    }

    private void insertUserImage() {
        btnInsertImage.startAnimation(scaleUpAnim);
        Intent intent = new Intent();
        intent.setType("image/*");
        intent.setAction(Intent.ACTION_GET_CONTENT);
        startActivityForResult(Intent.createChooser(intent, "Select an image"), 101);
    }

    @Override
    protected void onActivityResult(int requestCode, int resultCode, Intent data) {
        super.onActivityResult(requestCode, resultCode, data);
        if (resultCode == RESULT_OK && data != null){
            uriImage = data.getData();
            ivPicture.setImageURI(uriImage);
            imagePath = uriImage.getPath();
            imageSelected = true;

            Animation anim = AnimationUtils.loadAnimation(getApplicationContext(),
R.anim.fade_in_anim);
            ivPicture.setAnimation(anim);
        }
    }
}

```

The songs used within this project are the songs from YouTube Studio's audio library:

- Song1 – Event Horizon
- Song2 – On The Flip
- Song 3 – In The Morning
- Song4 – Purple Desire

activity_gallery.xml file:

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".Gallery">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:padding="10dp"

```

```

        android:text="TripBuddy Memory Gallery"
        android:textColor="@color/white"
        android:textAlignment="center"
        android:textSize="20dp"
        android:textStyle="bold"
        android:background="@color/cardview_dark_background"
    />

```

```

<GridLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:rowCount="6"
    android:columnCount="3"
    android:layout_centerInParent="true"
>
    <ImageView
        android:layout_marginTop="120dp"
        android:layout_height="120dp"
        android:layout_width="120dp"
        android:layout_margin="10dp"
        android:id="@+id/imgMemory1"
    />
    <ImageView
        android:layout_marginTop="120dp"
        android:layout_height="120dp"
        android:layout_width="120dp"
        android:layout_margin="10dp"
        android:id="@+id/imgMemory2"
    />

    <ImageView
        android:layout_marginTop="120dp"
        android:layout_height="120dp"
        android:layout_width="120dp"
        android:layout_margin="10dp"
        android:id="@+id/imgMemory3"
    />
    <ImageView
        android:layout_height="120dp"
        android:layout_width="120dp"
        android:layout_margin="10dp"
        android:id="@+id/imgMemory4"
    />

    <ImageView
        android:layout_height="120dp"
        android:layout_width="120dp"
        android:layout_margin="10dp"
        android:id="@+id/imgMemory5"
    />
    <ImageView
        android:layout_height="120dp"
        android:layout_width="120dp"
        android:layout_margin="10dp"
        android:id="@+id/imgMemory6"
    />

```

```

</GridLayout>

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="70dp"
    android:text="Description here"
    android:gravity="center"
    android:background="@color/black"
    android:textColor="@color/white"
    android:id="@+id/txtDescription"
/>

<Button
    android:text="Return to Home Dashboard"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_alignParentBottom="true"
    android:layout_centerInParent="true"
    android:id="@+id/btnReturnHome"
/>

<Button
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:id="@+id/btnPlayMusic"
    android:layout_alignParentBottom="true"
    android:layout_alignParentLeft="true"
    android:layout_marginBottom="60dp"
    android:text="Play" />

<Button
    android:text="Pause"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignParentBottom="true"
    android:layout_centerInParent="true"
    android:layout_marginBottom="60dp"
    android:id="@+id/btnPauseMusic"
/>

<Button
    android:text="Stop"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignParentBottom="true"
    android:id="@+id/btnStopMusic"
    android:layout_marginBottom="60dp"
    android:layout_alignParentRight="true"
/>

</RelativeLayout>

```


Gallery.java file:

```
package com.example.tripbuddy;

import android.content.Context;
import android.content.Intent;
import android.database.Cursor;
import android.media.MediaPlayer;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;
import android.widget.ImageView;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Gallery extends AppCompatActivity {
    MemoryDetails_SQLite dbMemoryDetails;

    private Button btnPlayMusic, btnPauseMusic, btnStopMusic, btnReturnHome;
    private MediaPlayer mediaPlayer;
    private ImageView imgMemory1, imgMemory2, imgMemory3, imgMemory4, imgMemory5,
imgMemory6;
    private TextView txtDescription;
    private Integer songName;
    public int[] arrMemoryID = new int[6];

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        dbMemoryDetails = new MemoryDetails_SQLite(this);
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_gallery);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets)
-> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });
        txtDescription = findViewById(R.id.txtDescription);
        btnReturnHome = findViewById(R.id.btnReturnHome);
        imgMemory1 = findViewById(R.id.imgMemory1);
        imgMemory2 = findViewById(R.id.imgMemory2);
        imgMemory3 = findViewById(R.id.imgMemory3);
        imgMemory4 = findViewById(R.id.imgMemory4);
```

```

imgMemory5 = findViewById(R.id.imgMemory5);
imgMemory6 = findViewById(R.id.imgMemory6);
btnPlayMusic = findViewById(R.id.btnPlayMusic);
btnPauseMusic = findViewById(R.id.btnPauseMusic);
btnStopMusic = findViewById(R.id.btnStopMusic);
txtDescription.setVisibility(View.INVISIBLE);

Cursor cursor = dbMemoryDetails.getAllMemoryEntries();
int memoryNum = 1;
if (cursor.moveToFirst()) {
    do {
        if (cursor.getString(1).equals(CurrentUserDetails.username)) {
            arrMemoryID[memoryNum - 1] = cursor.getInt(0);
            if (memoryNum == 1){
                imgMemory1.setImageURI(Uri.parse(cursor.getString(3)));
            }
            if (memoryNum == 2){
                imgMemory2.setImageURI(Uri.parse(cursor.getString(3)));
            }
            if (memoryNum == 3){
                imgMemory3.setImageURI(Uri.parse(cursor.getString(3)));
            }
            if (memoryNum == 4){
                imgMemory4.setImageURI(Uri.parse(cursor.getString(3)));
            }
            if (memoryNum == 5){
                imgMemory5.setImageURI(Uri.parse(cursor.getString(3)));
            }
            if (memoryNum == 6){
                imgMemory6.setImageURI(Uri.parse(cursor.getString(3)));
            }
            memoryNum++;
        }
    } while (cursor.moveToNext());
}
else{
    Toast.makeText(this, "Error! No memories found.",
Toast.LENGTH_SHORT).show();
}
cursor.close();

btnReturnHome.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        returnHome();
    }
});

imgMemory1.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        ViewGroup.MarginLayoutParams marginParams =
(ViewGroup.MarginLayoutParams) imgMemory1.getLayoutParams();
        marginParams.setMargins(0, 0, 0, 0);
        imgMemory1.setLayoutParams(marginParams);
    }
});

```

```

        ViewGroup.LayoutParams layoutParams = imgMemory1.getLayoutParams();
        layoutParams.width = ViewGroup.LayoutParams.MATCH_PARENT;
        layoutParams.height = ViewGroup.LayoutParams.MATCH_PARENT;
        imgMemory1.setLayoutParams(layoutParams);
        imgMemory1.setScaleType(ImageView.ScaleType.FIT_XY);
        loadImageDetails(1);
    }
});

imgMemory2.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        ViewGroup.MarginLayoutParams marginParams =
(ViewGroupGroup.MarginLayoutParams) imgMemory2.getLayoutParams();
        marginParams.setMargins(0, 0, 0, 0);
        imgMemory2.setLayoutParams(marginParams);

        ViewGroup.LayoutParams layoutParams = imgMemory2.getLayoutParams();
        layoutParams.width = ViewGroup.LayoutParams.MATCH_PARENT;
        layoutParams.height = ViewGroup.LayoutParams.MATCH_PARENT;
        imgMemory2.setLayoutParams(layoutParams);
        imgMemory2.setScaleType(ImageView.ScaleType.FIT_XY);
        loadImageDetails(2);
    }
});

imgMemory3.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        ViewGroup.MarginLayoutParams marginParams =
(ViewGroupGroup.MarginLayoutParams) imgMemory3.getLayoutParams();
        marginParams.setMargins(0, 0, 0, 0);
        imgMemory3.setLayoutParams(marginParams);

        ViewGroup.LayoutParams layoutParams = imgMemory3.getLayoutParams();
        layoutParams.width = ViewGroup.LayoutParams.MATCH_PARENT;
        layoutParams.height = ViewGroup.LayoutParams.MATCH_PARENT;
        imgMemory3.setLayoutParams(layoutParams);
        imgMemory3.setScaleType(ImageView.ScaleType.FIT_XY);
        loadImageDetails(3);
    }
});

imgMemory4.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        ViewGroup.MarginLayoutParams marginParams =
(ViewGroupGroup.MarginLayoutParams) imgMemory4.getLayoutParams();
        marginParams.setMargins(0, 0, 0, 0);
        imgMemory4.setLayoutParams(marginParams);

        ViewGroup.LayoutParams layoutParams = imgMemory4.getLayoutParams();
        layoutParams.width = ViewGroup.LayoutParams.MATCH_PARENT;
        layoutParams.height = ViewGroup.LayoutParams.MATCH_PARENT;
        imgMemory4.setLayoutParams(layoutParams);
        imgMemory4.setScaleType(ImageView.ScaleType.FIT_XY);

```

```

        loadImageDetails(4);
    }
});

imgMemory5.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        ViewGroup.MarginLayoutParams marginParams =
(ViewGroup.MarginLayoutParams) imgMemory5.getLayoutParams();
        marginParams.setMargins(0, 0, 0, 0);
        imgMemory5.setLayoutParams(marginParams);

        ViewGroup.LayoutParams layoutParams = imgMemory5.getLayoutParams();
        layoutParams.width = ViewGroup.LayoutParams.MATCH_PARENT;
        layoutParams.height = ViewGroup.LayoutParams.MATCH_PARENT;
        imgMemory5.setLayoutParams(layoutParams);
        imgMemory5.setScaleType(ImageView.ScaleType.FIT_XY);
        loadImageDetails(5);
    }
});

imgMemory6.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        ViewGroup.MarginLayoutParams marginParams =
(ViewGroup.MarginLayoutParams) imgMemory6.getLayoutParams();
        marginParams.setMargins(0, 0, 0, 0);
        imgMemory6.setLayoutParams(marginParams);

        ViewGroup.LayoutParams layoutParams = imgMemory6.getLayoutParams();
        layoutParams.width = ViewGroup.LayoutParams.MATCH_PARENT;
        layoutParams.height = ViewGroup.LayoutParams.MATCH_PARENT;
        imgMemory6.setLayoutParams(layoutParams);
        imgMemory6.setScaleType(ImageView.ScaleType.FIT_XY);
        loadImageDetails(6);
    }
});

btnStopMusic.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        if (mediaPlayer.isPlaying()) {
            mediaPlayer.stop();
            mediaPlayer = MediaPlayer.create(Gallery.this, songName);
        }
    }
});

btnPauseMusic.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        if (mediaPlayer.isPlaying()) {
            mediaPlayer.pause();
        }
    }
});

```

```

        btnPlayMusic.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                if (!mediaPlayer.isPlaying()) {
                    mediaPlayer.start();
                }
            }
        });
    }

    private void returnHome() {
        Intent intent = new Intent(Gallery.this, HomeDashboard.class);
        startActivity(intent);
    }

    private void loadImageDetails(int imageNum) {
        Cursor cursor = dbMemoryDetails.getAllMemoryEntries();

        if (cursor.moveToFirst()) {
            do {
                int originalID = cursor.getInt(0);
                if (cursor.getString(1).equals(CurrentUserDetails.username) &&
originalID == arrMemoryID[imageNum - 1]) {
                    txtDescription.setText(cursor.getString(2));
                    txtDescription.setVisibility(View.VISIBLE);
                    songName = cursor.getInt(4);
                    mediaPlayer = MediaPlayer.create(Gallery.this, songName);
                    mediaPlayer.start();
                }
            } while (cursor.moveToNext());
        }
        else{
            Toast.makeText(this, "Error! No memory found.", Toast.LENGTH_SHORT).show();
        }
        cursor.close();
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        if (mediaPlayer != null){
            mediaPlayer.release();
            mediaPlayer = null;
        }
    }
}

```

Deliverable 5

CurrentUserDetails.java class file:

```
package com.example.tripbuddy;

public class CurrentUserDetails {
    public static final String PREFS_NAME = "user_account_details";
    public static final String KEY_USERNAME = "username";
    public static final String KEY_TRIPAMOUNT = "total_trips";
    public static final String KEY_LOGGEDIN = "logged_in";
    public static final String KEY_DARKMODE = "dark_mode";
    public static final String KEY_BACKGROUNDMUSIC = "background_music";
    public static boolean loggedIn = false;
    public static String username = "";
    public static Integer totalTripsPlanned = 0;
    public static boolean darkMode = false;

    public static void populateUserDetails(String userName, Integer totalTrips, boolean
dark){
        loggedIn = true;
        username = userName;
        totalTripsPlanned = totalTrips;
        darkMode = dark;
    }
    public static void userLogOff(){
        loggedIn = false;
        username = "";
        totalTripsPlanned = 0;
        darkMode = false;
    }
}
```

TripDetails_SQLite.java class file:

```
package com.example.tripbuddy;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class TripDetails_SQLite extends SQLiteOpenHelper {
    private static final String DATABASE_NAME = "trip.db";
    private static final int DATABASE_VERSION = 1;
    public static final String TABLE_DETAILS = "details";
    public static final String KEY_ID = "trip_id";
    public static final String KEY_USERNAME = "username";
    public static final String KEY_DESTINATION = "destination";
    public static final String KEY_ACTIVITIES = "activities";
    public static final String KEY_CUSTOMEXPENSES = "custom_expense";
```

```

public TripDetails_SQLite(Context context) {
    super(context, DATABASE_NAME, null, DATABASE_VERSION);
}

@Override
public void onCreate(SQLiteDatabase db) {
    String CREATE_TABLE = "CREATE TABLE " + TABLE_DETAILS + "("
        + KEY_ID + " INTEGER PRIMARY KEY AUTOINCREMENT,"
        + KEY_USERNAME + " TEXT,"
        + KEY_DESTINATION + " TEXT,"
        + KEY_ACTIVITIES + " TEXT,"
        + KEY_CUSTOMEXPENSES + " DOUBLE" + ")";
    db.execSQL(CREATE_TABLE);
}

@Override
public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
    db.execSQL("DROP TABLE IF EXISTS " + TABLE_DETAILS);
    onCreate(db);
}

public boolean addNewTrip(String destination, String activities, double customEx) {
    SQLiteDatabase db = this.getWritableDatabase();
    ContentValues values = new ContentValues();
    values.put(KEY_USERNAME, CurrentUserDetails.username);
    values.put(KEY_DESTINATION, destination);
    values.put(KEY_ACTIVITIES, activities);
    values.put(KEY_CUSTOMEXPENSES, customEx);
    long num = db.insert(TABLE_DETAILS, null, values);
    db.close();
    if (num == -1){
        return false;
    }
    else{
        return true;
    }
}

public Cursor getAllTripEntries() {
    SQLiteDatabase db = this.getReadableDatabase();
    return db.rawQuery("SELECT * FROM " + TABLE_DETAILS, null);
}

public void deleteAllTripEntries() {
    SQLiteDatabase db = this.getWritableDatabase();
    db.delete(TABLE_DETAILS, null, null);
    db.close();
}
}

```

activity_user_login_screen.xml file:

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"

```

```

xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:padding="16dp"
tools:context=".UserLoginScreen">

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:padding="10dp"
    android:text="TripBuddy User Login"
    android:textColor="@color/white"
    android:textAlignment="center"
    android:textSize="20dp"
    android:textStyle="bold"
    android:background="@color/cardview_dark_background"
/>

<RelativeLayout
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_centerInParent="true"
    >
    <EditText
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center"
        android:hint="Enter username:"
        android:id="@+id/edtUsername"
        />

    <Button
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_centerHorizontal="true"
        android:text="Login"
        android:layout_marginTop="50dp"
        android:id="@+id/btnLogin"
        />
</RelativeLayout>
</RelativeLayout>

```

UserLoginScreen.java file:

```

package com.example.tripbuddy;

import android.content.Intent;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.view.View;
import android.view.animation.Animation;

```



```

import android.view.animation.AnimationUtils;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class UserLoginScreen extends AppCompatActivity {
    private Button btnLogin;
    private EditText edtUsername;
    Animation scaleUpAnim;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_user_login_screen);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets)
-> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });
        btnLogin = findViewById(R.id.btnLogin);
        edtUsername = findViewById(R.id.edtUsername);
        scaleUpAnim = AnimationUtils.loadAnimation(getApplicationContext(),
R.anim.size_up_anim);

        btnLogin.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                loadUserAccount();
            }
        });
    }

    private void loadUserAccount() {
        btnLogin.startAnimation(scaleUpAnim);
        try {
            String username = edtUsername.getText().toString();

            if(username.isEmpty() || username.isBlank()){
                Toast.makeText(this, "Username can't be left blank!",
Toast.LENGTH_SHORT).show();
            }
            else{
                SharedPreferences prefs =
getSharedPreferences(CurrentUserDetails.PREFS_NAME, MODE_PRIVATE);
                SharedPreferences.Editor editor = prefs.edit();
                if(prefs.contains(username)){

```

```

CurrentUserDetails.populateUserDetails(prefs.getString(CurrentUserDetails.KEY_USERNAME,
""), prefs.getInt(CurrentUserDetails.KEY_TRIPAMOUNT, 0),
prefs.getBoolean(CurrentUserDetails.KEY_DARKMODE, false));
    }
    else {
        editor.putString(CurrentUserDetails.KEY_USERNAME, username);
        editor.putInt(CurrentUserDetails.KEY_TRIPAMOUNT, 0);
        editor.putBoolean(CurrentUserDetails.KEY_LOGGEDIN, true);
        editor.putBoolean(CurrentUserDetails.KEY_DARKMODE, false);
        editor.putBoolean(CurrentUserDetails.KEY_BACKGROUNDMUSIC, true);
        editor.apply();
        CurrentUserDetails.populateUserDetails(username, 0, false);
    }
    Intent intent = new Intent(UserLoginScreen.this, HomeDashboard.class);
    startActivity(intent);
}
}
catch(Exception e){
    Toast.makeText(this, "Invalid username!", Toast.LENGTH_SHORT).show();
}
}
}

```

activity_settings_screen.xml file:

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".SettingsScreen">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:padding="10dp"
        android:text="TripBuddy Settings"
        android:textColor="@color/white"
        android:textAlignment="center"
        android:textSize="20dp"
        android:textStyle="bold"
        android:background="@color/cardview_dark_background"
    />

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Darke mode:"
        android:id="@+id/txtDarkMode"
        android:layout_centerInParent="true"
        android:textSize="20dp"
    />

```

```

        />

        <Switch
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_alignRight="@+id/txtDarkMode"
            android:layout_centerInParent="true"
            android:checked="false"
            android:id="@+id/swDarkMode"
        />

        <Button
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_alignParentBottom="true"
            android:text="Return to Home Dashboard"
            android:id="@+id/btnReturnToDashboard"
        />
    </RelativeLayout>

```

SettingsScreen.java file:

```
package com.example.tripbuddy;

import android.content.Intent;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.view.View;
import android.view.animation.Animation;
import android.view.animation.AnimationUtils;
import android.widget.Button;
import android.widget.CompoundButton;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.Switch;
import android.widget.ToggleButton;

import androidx.activity.EdgeToEdge;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class SettingsScreen extends AppCompatActivity {
    private Button btnReturnToDashboard;
    private Switch swDarkMode;
    Animation scaleUpAnim;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_settings_screen);
```

```

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets)
-> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });
        scaleUpAnim = AnimationUtils.loadAnimation(getApplicationContext(),
R.anim.size_up_anim);
        btnReturnToDashboard = findViewById(R.id.btnReturnToDashboard);
        swDarkMode = findViewById(R.id.swDarkMode);

        if (CurrentUserDetails.darkMode == true){
            swDarkMode.setChecked(true);
        }
        else{
            swDarkMode.setChecked(false);
        }

        swDarkMode.setOnCheckedChangeListener(new
CompoundButton.OnCheckedChangeListener() { (Coding, 2020)
            @Override
            public void onCheckedChanged(@NonNull CompoundButton buttonView, boolean
isChecked) {
                if (isChecked == true){
                    saveSettingChanges(true);
AppCompatActivity.setDefaultNightMode(AppCompatActivity.MODE_NIGHT_YES);
                }
                else if (isChecked == false){
                    saveSettingChanges(false);
AppCompatActivity.setDefaultNightMode(AppCompatActivity.MODE_NIGHT_NO);
                }
            }
        });

        btnReturnToDashboard.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                returnHome();
            }
        });
    }

    private void saveSettingChanges(boolean darkTheme) {
        SharedPreferences prefs = getSharedPreferences(CurrentUserDetails.PREFS_NAME,
MODE_PRIVATE);
        SharedPreferences.Editor editor = prefs.edit();

        CurrentUserDetails.darkMode = darkTheme;
        editor.putBoolean(CurrentUserDetails.KEY_DARKMODE, darkTheme);
        editor.apply();
    }

    private void returnHome() {

```

```
        btnReturnToDashboard.startAnimation(scaleUpAnim);  
        Intent intent = new Intent(SettingsScreen.this, HomeDashboard.class);  
        startActivity(intent);  
    }  
}
```

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